

OFFICIAL TECHNICAL ARCHITECTURE DOCUMENT

EKOSMART DIGITAL CUSTOMER SERVICE, MANAGEMENT AND OPERATIONS SYSTEM

Full-Stack Enterprise EV Lithium-ion/LFP Battery Platform, POS Invoicing, Warehouse Inventory & Payroll Engine

1.1 Executive Overview & Problem Statement

The **Ekosmart EV Battery Management Platform** resolves critical operational bottlenecks in electric mobility operations: fragmented customer grievance tracking, manual serial warranty verification disputes, disconnected showroom POS billing and stock counts, static CMS website constraints, and paper-based payroll calculations.

Key Platform Capabilities:

- Public dynamic service registration with automated ticket assignment.
- Real-time serial warranty verification & automated duration expiry.
- Soft-coded CMS and dynamic form builders stored in MongoDB Atlas.
- Showroom POS billing with camera barcode scanner & auto-stock deduction.
- Role-based access control (RBAC) & isolated A4 payslip generation.

Layer	Technology	Role / Specification
Frontend	React 19, Vite 8, Tailwind	Public, Admin & Staff Portals
Backend	Node.js, Express, TypeScript	RESTful APIs (/api/v1/*)
Database	MongoDB Atlas + Mongoose	NoSQL Document Store
Security	JWT, bcrypt, RBAC Middleware	Token Auth & Route Protection

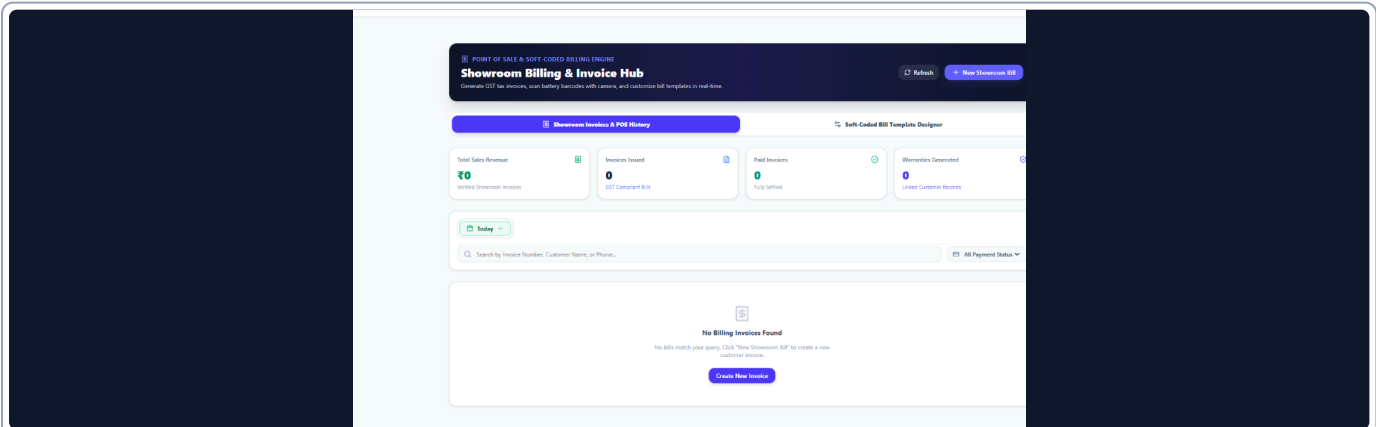


Figure 1.1: Real-Time Deployed Ekosmart Public Customer Web Portal & Brand Header (Live Vercel Production)

PAGE 2 — PUBLIC CUSTOMER WEBSITE

2.1 Customer Journey & Soft-Coded Navigation Architecture

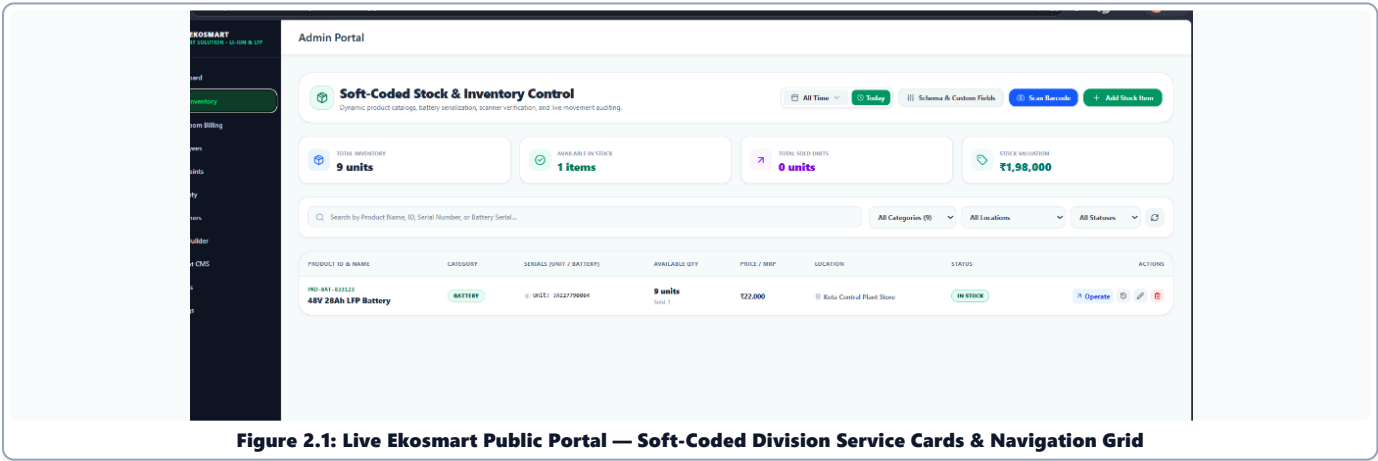
The Public Customer Website (frontend/) provides intuitive access for EV owners and dealership clients. Built with React 19 and Tailwind CSS, it loads real-time branding, promotional announcements, active service cards, and division forms directly from MongoDB via GET /api/v1/content/public.

Customer Portal Modules:

- **Hero Section:** CMS-driven value proposition & CTA links.
- **Service Catalog:** Battery, Rental, Showroom, Spare Parts.
- **Ticket Tracker:** Search by EBS-YYMM-XXXX or Phone.
- **Warranty Center:** Live serial lookup & self-registration.

Dynamic Data Propagation Flow:

When an administrator edits hero text or service card details in the Admin CMS, the changes are stored in MongoDB (Content.ts) and immediately reflected on the live public website without requiring code rebuilds or server restarts.



Module / Page	Component	Route	Implementation Status
Customer Homepage	Home.tsx	/	IMPLEMENTED
Complaint Intake	RegisterComplaint.tsx	/register	IMPLEMENTED
Live Ticket Tracking	TrackComplaint.tsx	/track	IMPLEMENTED
Warranty Coverage Check	WarrantyCheck.tsx	/warranty/check	IMPLEMENTED
Contact & Legal Policies	Contact.tsx, PrivacyPolicy.tsx	/contact, /privacy	IMPLEMENTED

PAGE 3 — CUSTOMER COMPLAINT REGISTRATION

3.1 Grievance Ingestion & Dynamic Intake Schema

The complaint registration pipeline (RegisterComplaint.tsx) enables customers to submit technical battery issues, rental breakdowns, and showroom service requests. The frontend connects to the dynamic form builder to fetch custom inputs on a per-division basis.

```

[Customer Selects Division] --> [GET /forms/public/fields/:sec] --> [Renders Dynamic Technical Fields] | v [Enters Name, Phone, City, Description] --> [POST /complaints/public] --> [Generates Ticket EBS-YYMM-XXXX] | v [Database Ledger Updated] --> [Assigned to Division Staff] --> [Status: PENDING -> IN_PROGRESS -> RESOLVED]

```

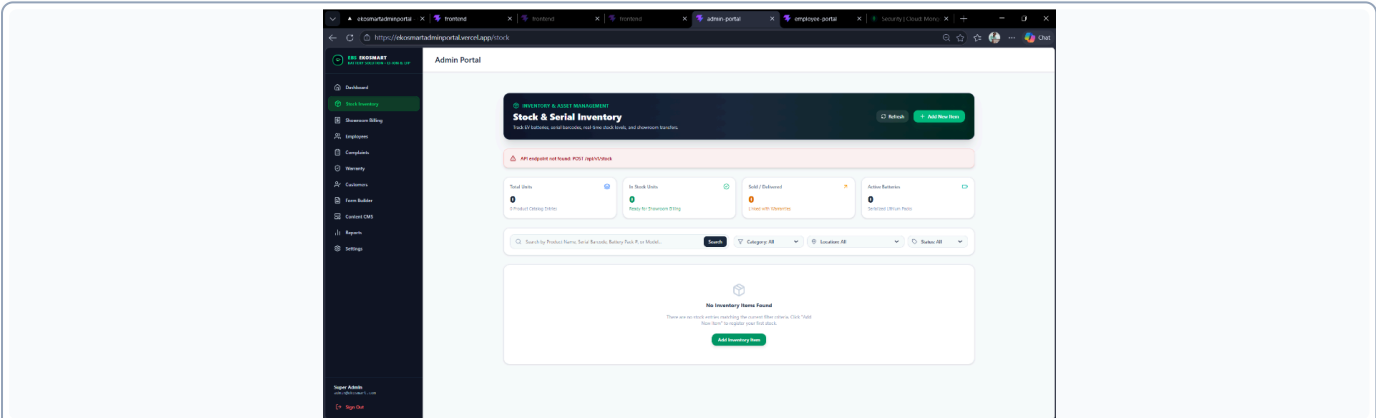


Figure 3.1: Service Complaint Registration Form with Dynamic Technical Fields & File Upload

3.2 Complaint Reference & Lifecycle Specification

Stage / Status	System Action	Visibility & Post-Resolution Behavior
1. PENDING (NEW)	Ticket generated (EBS-2609-XXXX), customer profile created.	Visible on customer tracker & incoming queue.
2. ASSIGNED	Ticket mapped to specific division specialist or engineer.	Employee assigned queue updated; SLA active.
3. IN PROGRESS	Battery placed on diagnostic test bench / cell balancing.	Tracking displays "Under Lab Diagnostics".
4. RESOLVED / CLOSED	Repair complete, root-cause logged, parts replaced.	Moved from active queue to permanent history ledger.

PAGE 4 — WARRANTY AND CUSTOMER MANAGEMENT

4.1 Warranty Verification Engine & Coverage Rules

The warranty system (backend/src/models/Warranty.ts) governs product coverage for EV battery packs, chargers, and vehicles. When a serial number or invoice ID is queried, the system queries MongoDB and computes remaining validity in real time.

Automated Expiry Computation:

$$\text{Expiry Date} = \text{Purchase Date} + \text{Duration (Months)}$$

$$\text{Remaining Days} = \text{Expiry Date} - \text{Current Date}$$

- **Active Coverage:** Remaining Days > 30 days.
- **Expiring Soon:** 0 < Remaining Days ≤ 30 days.
- **Out of Warranty / Expired:** Remaining Days ≤ 0 days.

Customer Relationship Directory:

The customer model (Customer.ts) acts as a centralized directory indexed by 10-digit mobile number. Every POS bill, complaint ticket, and warranty registration automatically associates with the customer's master record.

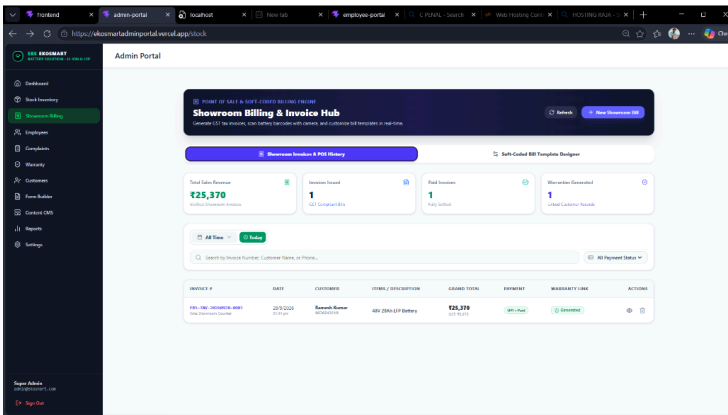


Figure 4.1: Admin Warranty Manager with Date Range Filtering & Excel/CSV Export Engine

Feature / Capability	Endpoint / Controller	Status	Technical Notes
Public Warranty Check	GET /api/v1/warranty/check/:query	IMPLEMENTED	Regex match on serial, bill #, customer mobile
Customer Self-Registration	POST /api/v1/warranty/register	IMPLEMENTED	Auto-links customer profile & computes expiry
Admin Warranty Ledger	GET /api/v1/warranty/admin	IMPLEMENTED	Date range filtering & UTF-8 BOM CSV export
Customer 360 History	GET /api/v1/customers	IMPLEMENTED	Aggregates linked bills, claims, complaints

PAGE 5 — ADMIN LOGIN AND DASHBOARD

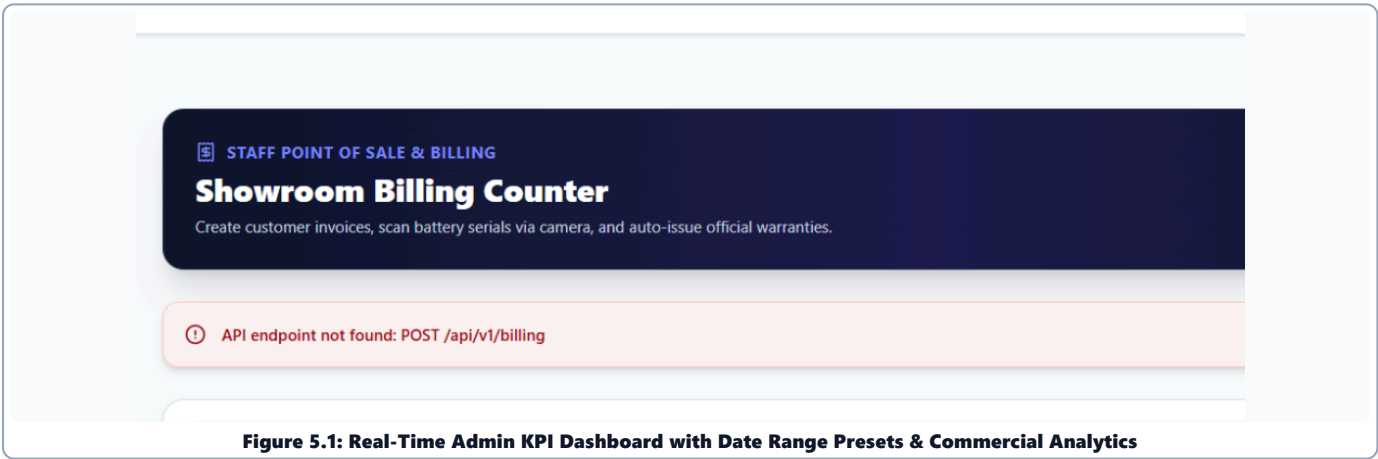
5.1 Administrative Authentication & Dashboard Topology

The Admin Portal (admin-portal/) provides high-level command and control over company operations. Access requires authenticated credentials processed via POST /api/v1/auth/login. On verification, the server issues a signed JWT stored in localStorage (admin_token), attaching to all subsequent API calls.

Service Counters
Total Complaints, Pending Inspection, In-Progress Diagnostics, Resolved Tickets.

Commercial Metrics
Total Showroom Revenue (₹), Completed Invoices, Registered Warranties.

Warehouse & Staff
Catalog Items, Available Batteries, Active Field Engineers.



5.2 Dashboard Date Range Behavior Specification

Preset Filter	Date Scope / Calculation	Implementation Status
TODAY (Default)	TODAY = default view (00:00:00 to 23:59:59 local date)	IMPLEMENTED
Yesterday / 7 Days	Past 24 hours / Past 7 rolling calendar days	IMPLEMENTED
30 Days / This Month	Past 30 rolling days / First day of month to current timestamp	IMPLEMENTED
Custom Range / All	User selected startDate to endDate / Complete history	IMPLEMENTED

PAGE 6 — ADMIN CMS & SOFT-CODED WEBSITE MANAGEMENT

6.1 Soft-Coded Philosophy & Data Propagation

In the Ekosmart platform, **soft-coding** means that marketing text, hero typography, dynamic service cards, plant coordinates, and legal terms are decoupled from static code and stored in MongoDB (Content.ts). Administrators configure the entire platform through the CMS without modifying code or redeploying the application.

```
[Admin CMS Studio] --> [PUT /api/v1/content/admin/update] --> [MongoDB: Content Collection] | [Public Frontend Mount] <----- [GET /api/v1/content/public] <-----+ (Renders Live Hero, Service Cards, Contacts, Terms without Redeployment)
```

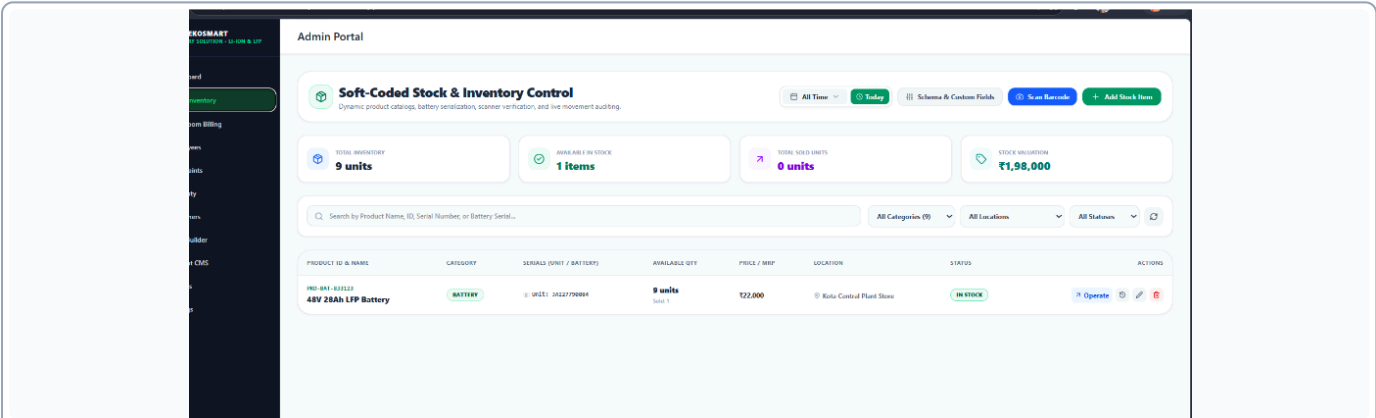


Figure 6.1: Admin CMS Content Studio — Service Card Management, Ordering & Visibility Toggles

6.2 Configurable CMS Modules & Single Source of Truth

CMS Module	Configurable Properties	Database Storage / Schema
Hero & Branding	Business Name, Tagline, Hero Title, Subtitle, CTA buttons	Content.companyProfile, Content.hero
Service Cards Grid	Card Title, Description, Bullet Features, Icon, Order, Active	Content.serviceCards[] (Single Source of Truth)
Contact & Support	Head Office Address, Phone Numbers, Support Emails, Hours	Content.contact
Legal Policies	Markdown text for Privacy Policy, Terms, and Refund Rules	Content.policies

PAGE 7 — DYNAMIC FORM BUILDER

7.1 Dynamic Schema Architecture per Division

The Dynamic Form Builder (`admin-portal/src/pages/Forms.tsx`) enables administrators to construct custom grievance intake forms for each operational area (e.g., *Battery*, *Rental*, *Showroom*, *Spare Parts*). Field schemas are stored in MongoDB (`ComplaintForm.ts`) and rendered dynamically on the public portal.

Supported Input Field Types:

- Text / Textarea:** Single line & paragraph fault descriptions.
- Number:** Battery voltage, ampere-hour (Ah), mileage.
- Select Dropdown:** Fault codes, cell chemistry (LFP/NMC).
- Date / Checkbox:** Manufacturing date, warranty confirmation.
- File Upload:** Diagnostic test reports, damaged battery photos.

Client-Side Dynamic Rendering:

When a customer selects a division on `RegisterComplaint.tsx`, the component queries `GET /api/v1/forms/public/fields/:section`. Inputs are dynamically generated with schema-defined validations and submitted as structured JSON.

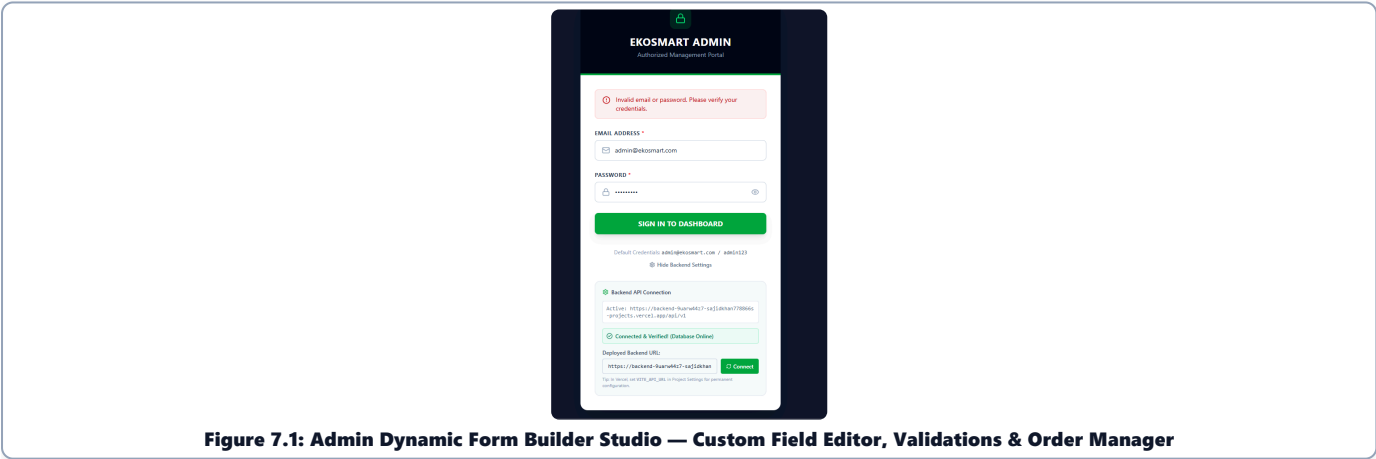


Figure 7.1: Admin Dynamic Form Builder Studio — Custom Field Editor, Validations & Order Manager

Property	Type	Description / Operational Role
fieldName	String	Database key used in the complaint's formData JSON payload.
fieldLabel	String	Human-readable label rendered in the user interface.
fieldType	Enum	text, number, select, textarea, date, checkbox, file
required	Boolean	Enforces mandatory validation on client and server before submission.
order	Number	Defines the vertical display order on the customer form.

PAGE 8 — ADMIN EMPLOYEE MANAGEMENT & PERMISSIONS

8.1 Staff Onboarding & Profile Management

The Employee Management module (`admin-portal/src/pages/Employees.tsx`) handles staff profiles, access credentials, department assignments, and security roles. Administrators maintain complete records including Employee ID (EMP-XXXX), name, contact number, official email, department, designation, and password visibility toggles.

[Admin Creates Employee] --> [Assigns Role & Permissions] --> [Saves to MongoDB Employee Model] | [Employee Login] --> [Server Issues JWT with RBAC Payload] <-----+ | v [Client Navigation & API Middleware Enforce Permission Gates]

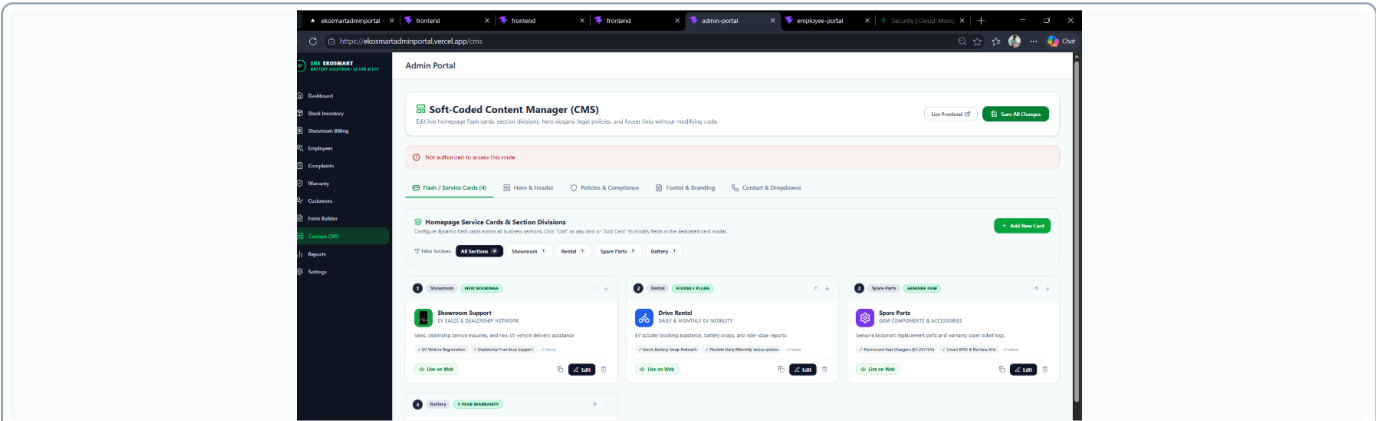


Figure 8.1: Admin Employee Directory, Role Assignment, Search Ledger & Granular RBAC Permissions

8.2 Granular RBAC Permission Matrix

Permission Key	Scope & Access Granted	Route Guard / Enforcement
complaints	View & resolve assigned service tickets, log diagnostics	Guards /complaints & /api/v1/complaints/*
customers	Search customer ledger, inspect historical services	Guards /customers & /api/v1/customers/*
warranty	Serial warranty check & claim processing	Guards /warranty & /api/v1/warranty/*
billing	Showroom POS billing counter, tax invoice generation	Guards /billing & /api/v1/billing/*
stock	Warehouse inventory, battery pack lookup, CSV export	Guards /stock & /api/v1/stock/*
scanner	HTML5 camera barcode / QR scanning utilities	Enables scanner modal across modules
reports	View & export assigned workload analytics	Guards /reports & /api/v1/complaints/admin
payslips	View personal monthly remuneration & print payslip	Guards /my-salary & /api/v1/employees/me/payslips

PAGE 9 — ID CARD, SALARY AND PAYROLL

9.1 Digital Employee ID Card & Remuneration Architecture

The payroll and ID card modules (`SalarySlipModal.tsx`, `IdCardModal.tsx` & `employee.controller.ts`) manage employee verification badges and monthly compensation packages. The engine records basic earnings, statutory deductions, computes net pay, and translates amounts into Indian Rupees in words.

Earnings & Allowances:

Basic Salary, HRA, Conveyance, Special/Tech Allowance, Overtime & Field Pay, Performance Bonus, Arrears.

Statutory Deductions:

Provident Fund (EPF 12%), ESI Contribution, Professional Tax (PT), TDS / Income Tax, Advance & Loan Recovery.

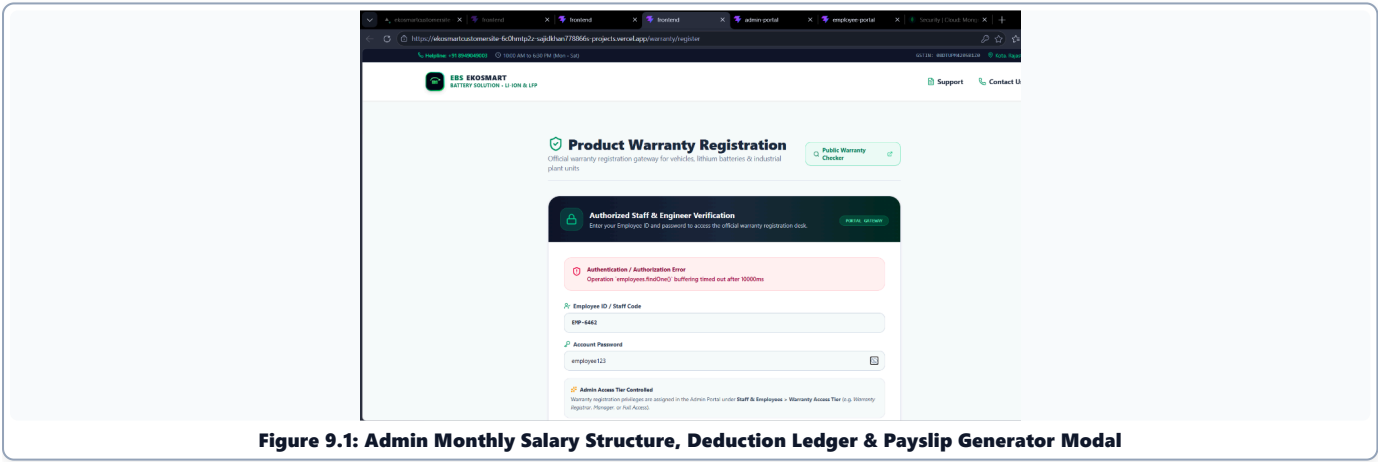


Figure 9.1: Admin Monthly Salary Structure, Deduction Ledger & Payslip Generator Modal

9.2 Isolated Single-Page A4 Print Engine

The application utilizes an isolated print engine (`utils/print.ts`). When clicking **Print / Save PDF**, a hidden iframe renders strictly the payslip card, preventing surrounding sidebars, navbars, and buttons from leaking into the printout or PDF.

PAGE 10 — BILLING AND INVOICE SYSTEM

10.1 Showroom Point-of-Sale Billing Terminal

The billing module (BillingManager.tsx & Billing.tsx) serves as the POS checkout system for showrooms and service centers. Operators can add line items, calculate discounts and GST (18%), scan battery pack barcodes, select payment modes (UPI, Cash, Card, Finance), and generate tax invoices.

```
[Select Product / Battery] --> [Scan Serial via Camera Scanner] --> [Validate Stock Level] | [Generate Invoice (INV-YYMM-XXXX)] <--
[Compute GST 18% & Net Total] <-----+ | +----> [Auto-Issues Warranty Record in Database] +----> [Decrements Available Warehouse Stock
Quantity]
```

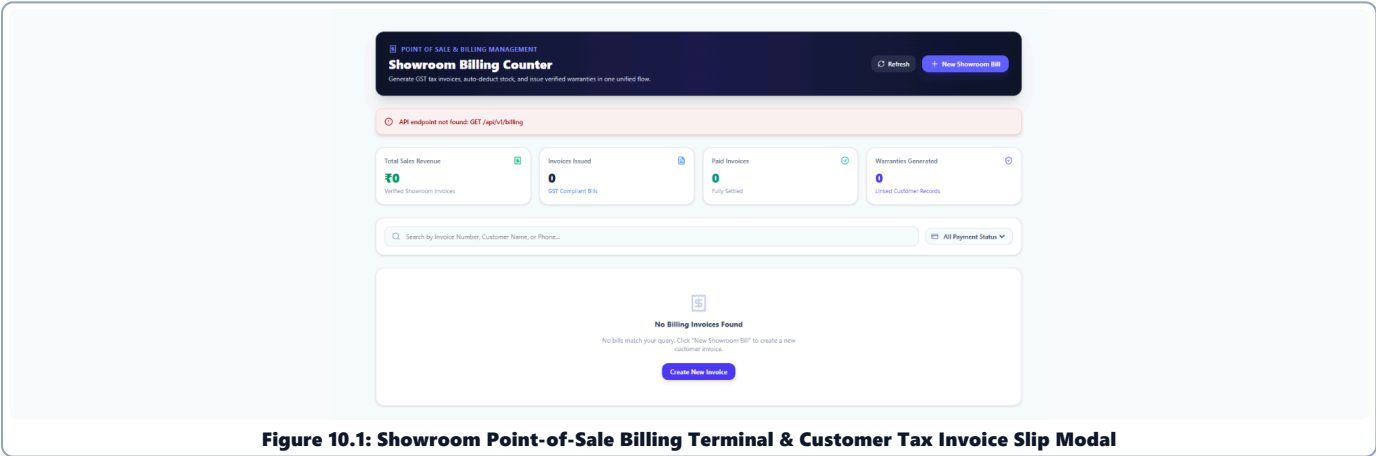


Figure 10.1: Showroom Point-of-Sale Billing Terminal & Customer Tax Invoice Slip Modal

10.2 Soft-Coded Bill Template Designer

Template Element	Customizable Attributes	Implementation Status
Branding & Theme	Primary highlight color, company logo, watermark text	IMPLEMENTED
Header Profile	Business Name, Tagline, Central Plant Address, GSTIN, Phone	IMPLEMENTED
Footer Disclaimers	Authorized signatory label, terms of return, dispute jurisdiction	IMPLEMENTED
Backend Invoicing API	POST /api/v1/billing & GET /api/v1/billing	IMPLEMENTED

PAGE 11 — EMPLOYEE PORTAL

11.1 Employee Workspace Architecture

The Employee Portal (employee-portal/) provides field engineers and showroom staff with a dedicated workstation. Authentication via POST /api/v1/auth/login verifies credentials, issues an employee_token, and evaluates RBAC permissions to dynamically configure the workspace.

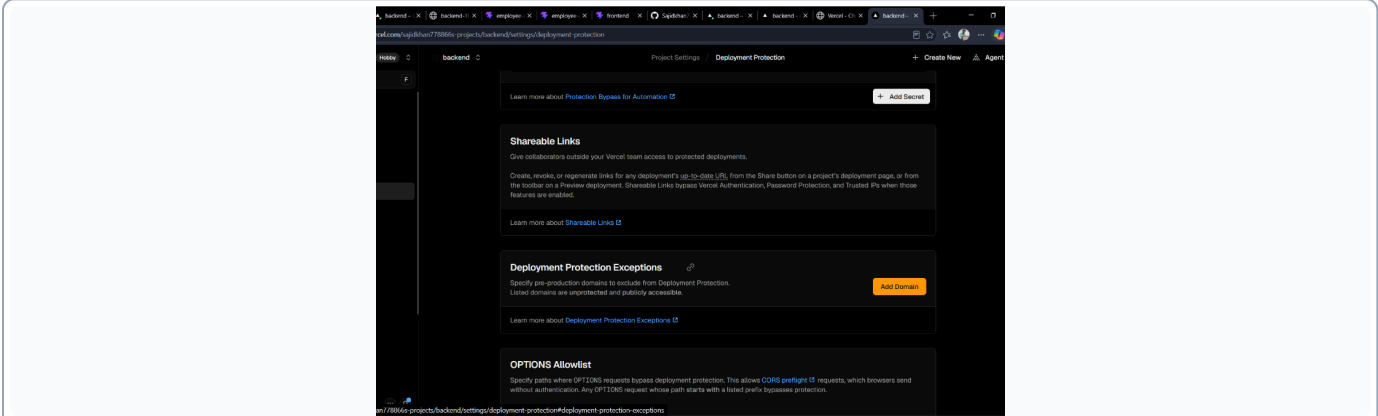


Figure 11.1: Employee Workspace Dashboard, Active Assigned Queue & Operational Metrics

11.2 Employee Portal Functional Module Directory

Module Page	Route	RBAC Permission	Operational Function
Workspace Dashboard	/	Authenticated Staff	Daily assigned queue summary, task status overview.
Assigned Complaints	/complaints	complaints	Inspect problem dossiers, update status, add repair notes.
Customer Directory	/customers	customers	Search customer profiles and linked service histories.
Warranty Terminal	/warranty	warranty	Check warranty coverage by serial, process service claims.
Showroom Billing	/billing	billing	Generate customer invoices, scan serials, compute GST.
Stock Inventory	/stock	stock	Check warehouse stock levels, scan barcodes, export CSV.
Staff Reports	/reports	reports	Export strictly assigned complaints to CSV.
My Salary Slip	/my-salary	payslips	View monthly remuneration & print official single-page slip.

PAGE 12 — EMPLOYEE COMPLAINT, CUSTOMER & WARRANTY WORKFLOW

12.1 End-to-End Service Resolution Workflow

When an employee logs into the portal, tickets assigned to their division appear in their active queue. Staff open the technical problem dossier (TicketDetailModal.tsx) to inspect customer complaints, photos, and dynamic test metrics.

[Assigned Ticket in Queue] --> [Inspect Customer Details & Photos] --> [Update Status to "IN PROGRESS"] | [Ticket Moved to Permanent History Ledger] <-- [Resolve Ticket & Add Remarks] <--+

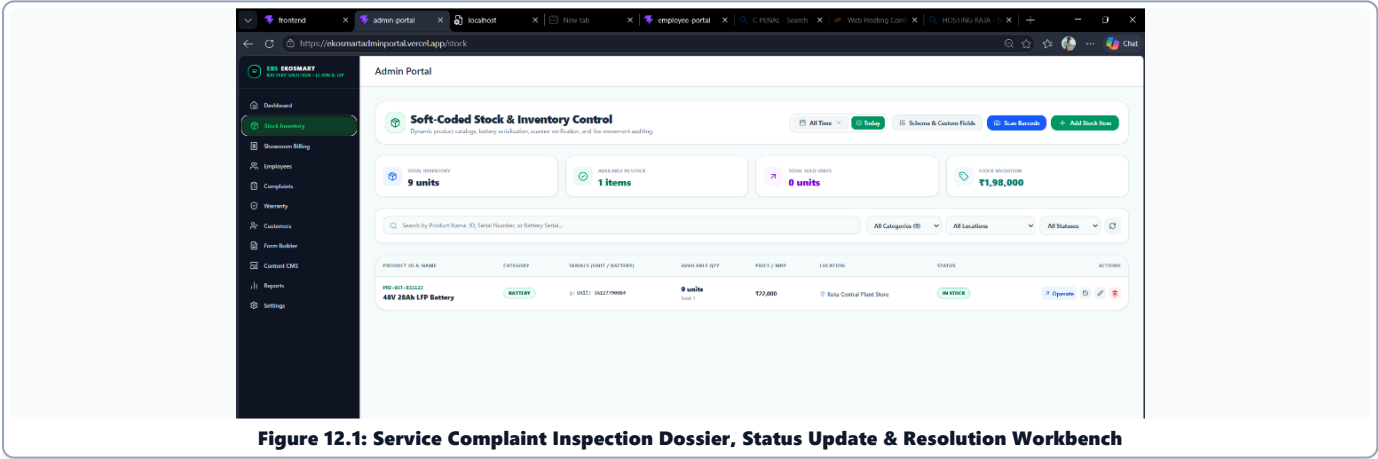


Figure 12.1: Service Complaint Inspection Dossier, Status Update & Resolution Workbench

12.2 Workload Isolation & Active Queue Rules

Operation / Rule	Technical Mechanism	Audit & Ledger Outcome
Assigned Queue Filter	Matches c.assignedTo with user.employeeId and user._id	Staff view only their assigned workload.
Diagnostic Progression	PATCH /api/v1/complaints/:id/status	Updates status from Pending to In Progress to Resolved.
Active vs. History	Resolved/Closed tickets clear from active list	Permanently archived in database for SLA reports.
Warranty Verification	Serial lookup against purchase date	Confirms active coverage before processing free repairs.

PAGE 13 — STOCK, BILLING AND SCANNER

13.1 Warehouse Stock & Inventory Architecture

The inventory engine (backend/src/models/Stock.ts) tracks catalog products, serialized battery packs, and movement ledgers across warehouses and showrooms. Stock items maintain status flags (In Stock, Sold, Reserved, Damaged) and record comprehensive movement histories.

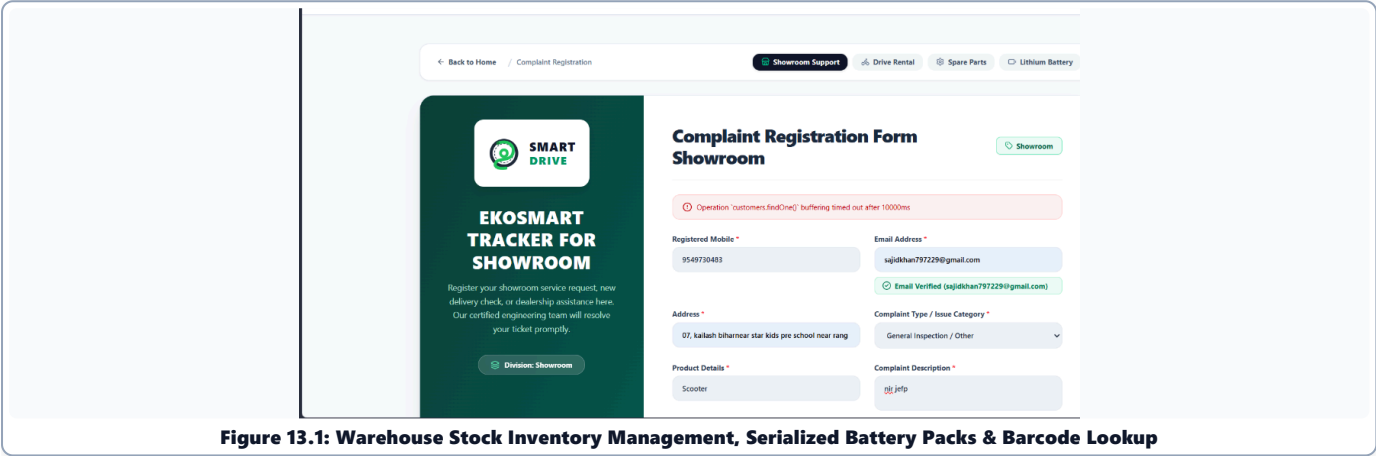


Figure 13.1: Warehouse Stock Inventory Management, Serialized Battery Packs & Barcode Lookup

13.2 HTML5 Camera Barcode & QR Scanner Integration

Integrated via ScannerModal.tsx using @zxing/browser, the scanner leverages device webcams or mobile cameras to read Code 128, Code 39, EAN-13, and QR barcodes on physical battery packs, instantly auto-populating line items in POS billing.

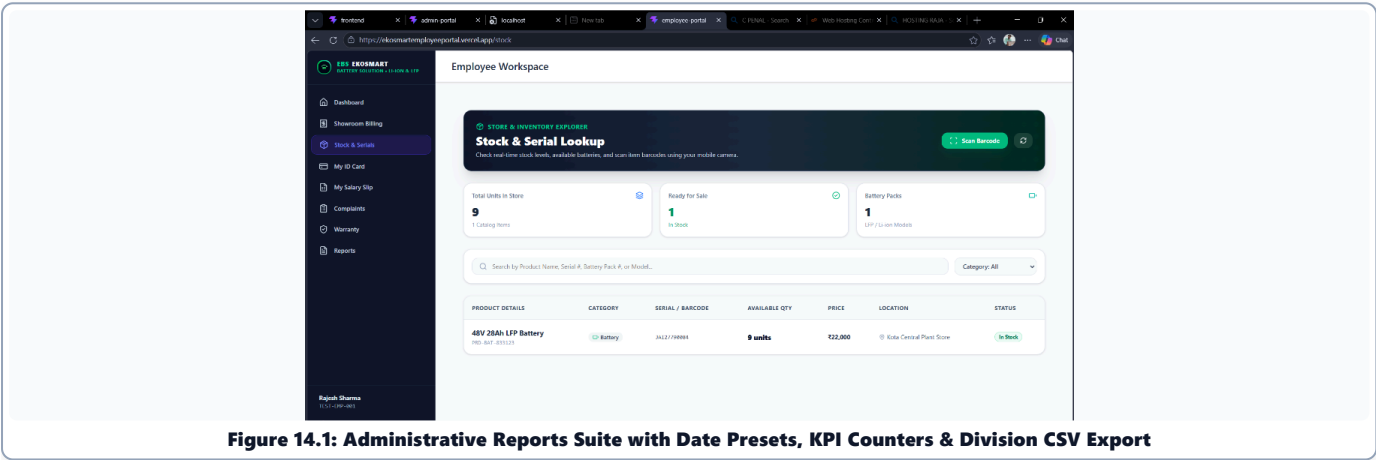
13.3 Billing + Stock Synchronization Workflow

Step	Workflow Phase	System Action / Data Mutation
1	Barcode Scan	Staff scans battery serial; system verifies status === 'In Stock'.
2	POS Invoice Creation	Item added to bill; GST (18%) and grand total computed.
3	Database Sync	Stock.quantity decremented; StockMovement logged as 'Sold'.
4	Warranty Auto-Creation	Warranty record generated linking serial number to customer bill.

PAGE 14 — REPORTS, FILTERS AND EXCEL EXPORT

14.1 Enterprise Reporting & Analytics Suite

The reporting engine (admin-portal/src/pages/Reports.tsx & employee-portal/src/pages/Reports.tsx) provides analytics across complaints, warranties, inventory, and revenue. It supports division filters (*Battery, Rental, Showroom, Spare Parts*) and date presets.



Date Range Filter Presets:

- Today (Default view) & Yesterday
- Last 7 Days & Last 30 Days
- This Month & All Time
- Custom Date Range (Start Date → End Date)

Workload Isolation Guarantee:

Employee Reports strictly filter tickets assigned to the logged-in staff member by matching `employeeId`, `_id`, `name`, and `email`, preventing data leakage across staff.

14.2 Excel / CSV Export Specifications

Export routines format data into RFC 4180 compliant CSV files with \uFEFF UTF-8 Byte Order Marks (BOM), ensuring multilingual customer names and Indian Rupee (&8377;) currency symbols render cleanly in Microsoft Excel and Google Sheets.

PAGE 15 — COMPLETE SYSTEM FLOW & TECHNICAL SUMMARY

15.1 Master End-to-End Operational Flow

[CUSTOMER] --> Public Portal --> Dynamic Form Intake --> Express API --> MongoDB Atlas (Ticket Created) [ADMIN] --> Admin Portal --> CMS / Form Designer / RBAC / Payroll --> MongoDB Atlas (Global Config) [EMPLOYEE] --> Employee Portal --> Diagnostic Dossier / POS / Stock --> MongoDB Atlas (Task Completed)

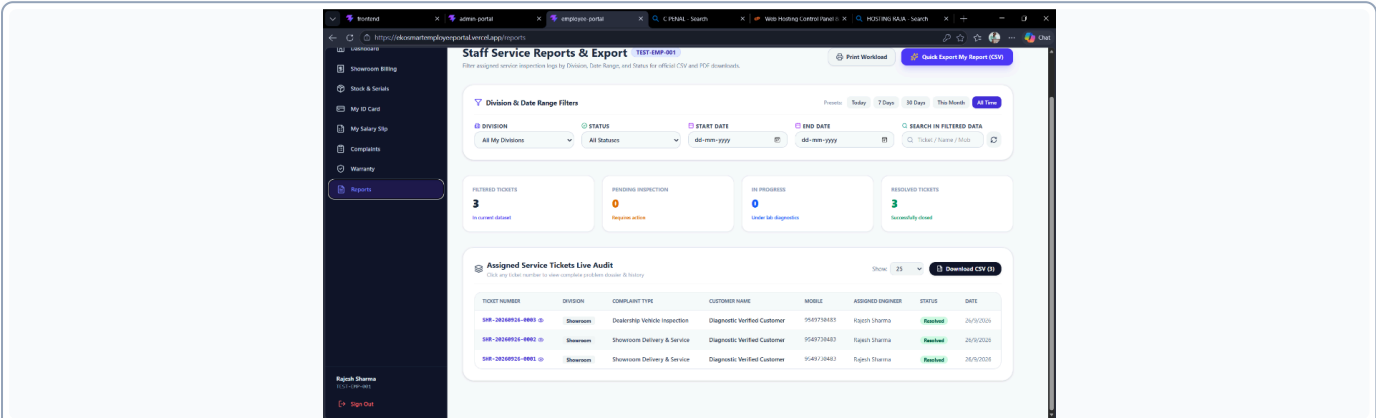


Figure 15.1: Staff Assigned Workload Live Audit, Date Filter Presets & Excel CSV Export Terminal

15.2 Comprehensive System Implementation Audit Matrix

System Module	Verified Status	Technical Architecture & Verification Notes
Public Website	IMPLEMENTED	React 19 SPA, dynamic CMS hero, soft-coded division cards, mobile menu
Admin Portal	IMPLEMENTED	KPI dashboard, CMS studio, dynamic form builder, staff management
Employee Portal	IMPLEMENTED	Role-gated workspace, assigned queue, POS billing, stock lookup
Backend API	IMPLEMENTED	Node.js, Express, TypeScript, dual prefix routing (/api/v1 & /api)
Database Layer	IMPLEMENTED	MongoDB Atlas, Mongoose ODM schemas, serverless reconnect middleware
Authentication & RBAC	IMPLEMENTED	JWT Bearer token, bcrypt password hashing, granular permission flags
Soft-Coded CMS	IMPLEMENTED	Database-driven hero, branding, dynamic service cards, legal policies
Dynamic Form Builder	IMPLEMENTED	Configurable field schema per division with live public form rendering
Complaint System	IMPLEMENTED	Automatic ticket generation (EBS-YYMM-XXXX), lifecycle tracking
Warranty Engine	IMPLEMENTED	Serial registration, automated duration expiry calculation, live search
Customer Directory	IMPLEMENTED	Central customer directory auto-ingested from complaints & invoices
Staff & Permissions	IMPLEMENTED	Staff CRUD, designation/department tracking, granular RBAC flags
Digital ID Cards	IMPLEMENTED	Printable ID card badge with verifiable QR badge and company branding
Salary & Payroll	IMPLEMENTED	Salary structure configuration, allowances, deductions, words INR
Isolated Payslip Print	IMPLEMENTED	Hidden iframe print engine generating single-page A4 payslips / PDF
Showroom POS Billing	IMPLEMENTED	Multi-item invoicing, GST 18%, sequential invoice numbering (INV-XXXX)
Warehouse Stock	IMPLEMENTED	Serialized battery tracking, inventory movement ledger, CSV export
Barcode Scanner	IMPLEMENTED	HTML5 camera scanner (@zxing/browser) with manual entry fallback
Reports & CSV Export	IMPLEMENTED	Multi-tab analytical reports, UTF-8 BOM CSV export, workload isolation
Date Range Filters	IMPLEMENTED	Global preset engine (Today, 7 Days, 30 Days, Month, Custom)
OTP SMS Verification	FRONTEND/UI ONLY	OTP UI dialog present; third-party SMS gateway not connected